COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 10 : SLO 1 — Tutorial: Restoring Division
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Problem 1: Restoring Division of 0111 (7) ÷ 0011 (3)

• Init: A = 0000, Q = 0111, M = 0011

• Step 1: shift (A,Q) → A = 0000, Q = 1110; A = A - M = 0000 - 0011 = 1101 (< 0) → restore A = 0000, Q0 = 0 → A = 0000, Q = 1110

• Step 2: shift → A = 0001, Q = 1100; A = A - M = 0001 - 0011 = 1110 (< 0) → restore A = 0001, Q0 = 0 → A = 0001, Q = 1100

• Step 3: shift → A = 0011, Q = 1000; A = A - M = 0011 - 0011 = 0000 (≥ 0) → Q0 = 1 → A = 0000, Q = 1001

• Step 4: shift → A = 0001, Q = 0010; A = A - M = 0001 - 0011 = 1110 (< 0) → restore A = 0001, Q0 = 0 → A = 0001, Q = 0010

• Final Quotient = Q = 0010 (binary) = 2 (decimal)

• Final Remainder = A = 0001 (binary) = 1 (decimal)

• Verification: 7 ÷ 3 = 2 remainder 1 ✓